

Decorator Calculator Program

1. Data Flow Diagram (DFD)

User → Enter numbers and choice → Decorator Calculator → Result Output

2. Algorithm

1. Start
2. Read two numbers from user
3. Display operation menu
4. Read user choice
5. If choice is Add, perform addition
6. If choice is Subtract, perform subtraction
7. If choice is Divide, check divisor is not zero and divide
8. Display result
9. Stop

3. Sample Output

Input:

Enter first number: 10

Enter second number: 5

Enter choice: 1

Output:

Addition is: 15

Code

```
# Decorator definition
def calculator(func):
    def wrapper(a, b, choice):
        if choice == 1:
            print("Addition is:", a + b)
        elif choice == 2:
            print("Subtraction is:", a - b)
        elif choice == 3:
            if b == 0:
                print("Cannot divide by zero")
            else:
                print("Division is:", a / b)
        else:
            print("Invalid choice")
        return func(a, b, choice)
    return wrapper

# Using the decorator
@calculator
def calculate(a, b, choice):
    pass

# User input
x = int(input("Enter first number: "))
y = int(input("Enter second number: "))

print("1. Add")
print("2. Subtract")
print("3. Divide")

ch = int(input("Enter your choice: "))
calculate(x, y, ch)
```

OUTPUT ⇒

```
Enter first number: 10
Enter second number: 5
1. Add
2. Subtract
3. Divide
Enter your choice: 1
Addition is: 15
```